

Supplementary materials for
The neural correlates of domain-general visual ability

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Selection of novel objects for the fMRI experiment.

Since o is domain-general, measuring neural sensitivity with many novel object domains should afford more construct coverage than selecting only a few domains. We chose objects that vary in attributes shown to engage differing neural substrates, including animacy (Noppeney et al. 2006; Wiggett et al. 2009), surface texture (Echavarria et al. 2016) and curvilinearity/rectilinearity (Nasr et al. 2014; Yue et al. 2014). Any region showing evidence of sensitivity to object identity in a set of novel domains with varying levels of these attributes is more likely to reflect attribute-general (and thus domain-general) neural correlates.

A set of novel objects from 29 categories was collected from several sources and included computer-generated novel object domains (Wong and Hayward 2005; Op de Beeck et al. 2008; Leek et al. 2012; Biederman et al. 2014; Horst and Hout 2016; Norman et al. 2016; Watson et al. 2017). Additional images were gathered from the Internet, including images of abstract three-dimensional sculptures, three-dimensional printed abstract chess pieces and dog toys. None of the three novel domains used in the behavioral study (vertical Ziggerins, Sheinbugs and symmetric Greebles) were used during scanning. All images were transformed to greyscale and equated for low-level images properties using the Matlab SHINE toolbox (Willenbockel et al. 2010). The algorithm was used to equate each pair of alternated – or identical but rotated - images in their Fourier amplitude and luminance histograms through 5 iterative repetitions (this was sufficient as the image pairs were highly similar to start with).

To select specific pairs of objects that varied in their similarity, we conducted an online pilot study in which we had participants perform same/different judgments on 3 pairs

of objects, using 3 objects from each of 29 categories. On each trial of a matching task, participants studied one object for 800 ms followed by a mask for 500 ms, and the second object. Participants decided whether the two objects were the same or not. On trials in which the correct response was same, the studied object was rotated 15 degrees in-plane clockwise so that the images did not match exactly.

There were 87 trials in which the correct response was different and 87 trials in which the correct response was same (one trial for each object). The order of the 174 trials was randomized. Data from 45 participants were collected via Amazon Mechanical Turk. Five participants were excluded for not following instructions, leaving 40 participants for analysis (29 male, mean age = 36.2 years). For each object pair, the correlation between subjects' average accuracy and item response were calculated, as well as the difference between hit rate and false alarm rate (HIT-FA) were calculated for each object pair. For each category we selected two pairs to use in the adaptation runs based on these values, trying to maximize the correlation with overall performance as well as a wide range of similarity for object pairs (as indexed by HIT-FA). We selected 52 pairs from 26 distinct categories, with a mean HIT-FA score was 0.44, with a SD of .31, skewness of -0.40 and kurtosis of -0.71. The reliability of the measurements gathered online for this subset of pairs was high (Cronbach alpha = 0.91), suggesting these object pairs tap into a sufficient range of visual ability.

Selection of participants for the fMRI experiment.

Participants were recruited from a previous behavioral study (Sunday et al. 2021) to sample a range of object recognition abilities. The histogram (Figure 1-1) shows the 282

participants from the behavioral study(Sunday et al. 2021), with the 37 MRI participants recruited for the current study indicated by red tick marks. One participant's data were eventually excluded because of large motion artifacts through the experimental runs.

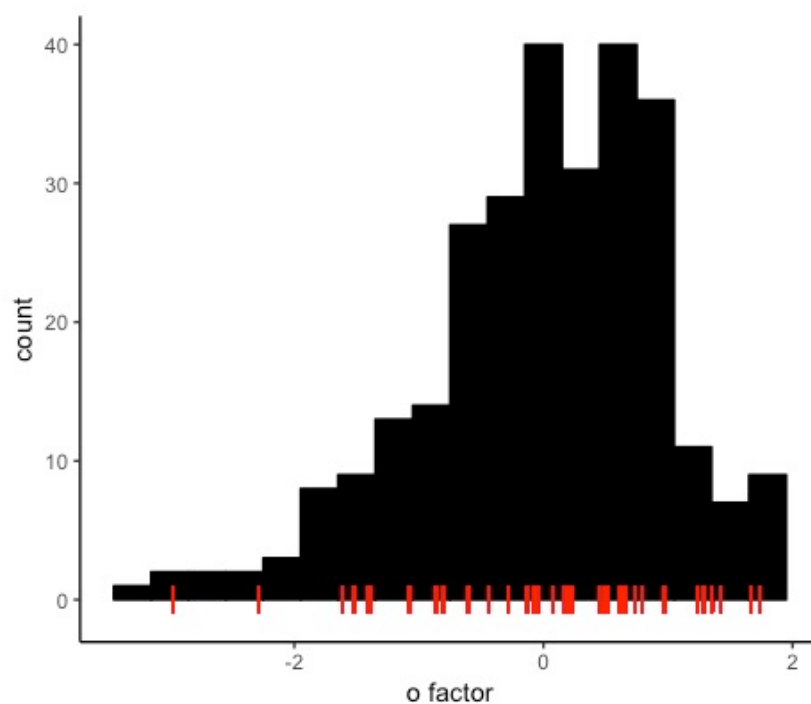


Figure S1. Histogram of o factor scores for all participants in(Sunday et al. 2021), indicating the level of ability of the 37 subjects recruited for the fMRI study.

Table S1

Mean MNI coordinates (with SD) for individually-defined ROIs.

ROI	X	Y	Z
IEBA	-35.4 (9.4)	-78.3 (4.5)	-2.8 (9.8)
IFBA1	-38.1 (6.5)	-60.0 (5.9)	-15.4 (4.3)
IFBA2	-34.6 (7.4)	-47.7 (6.7)	-16.7 (5.2)
IOFA	-35.4 (7.3)	-78.1 (8.5)	-11.1 (6.8)
IFFA1	-37.6 (5.7)	-60.1 (6.6)	-16.7 (3.5)
IFFA2	-35.3 (6.3)	-48.9 (8.6)	-18.0 (3.9)
ILOC	-41.2 (5.0)	-77.1 (6.5)	-2.6 (6.7)
IMOA1	-26.3 (4.0)	-61.4 (5.9)	-12.8 (2.9)
IMOA2	-26.8 (4.5)	-46.0 (5.7)	-13.2 (3.9)
rEBA	40.2 (9.1)	-73.3 (5.6)	-4.1 (9.5)
rFBA1	36.5 (7.8)	-56.6 (7.2)	-15.4 (4.1)
rFBA2	35.5 (7.6)	-44.1 (7.9)	-15.9 (5.8)
rOFA	37.4 (7.8)	-77.4 (7.0)	-9.5 (7.6)
rFFA1	35.7 (5.9)	-58.5 (6.2)	-14.6 (7.0)
rFFA2	35.6 (6.6)	-44.3 (6.6)	-18.1 (3.7)
rLOC	44.5 (5.0)	-73.3 (7.4)	-2.6 (6.8)
rMOA1	27.9 (4.2)	-61.2 (6.5)	-11.6 (4.0)
rMOA2	28.3 (4.0)	-44.4 (5.5)	-12.3 (6.4)

Table S2

Pearson Correlations for residual responses in different ROIs, with Bayesian support for a correlation

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	17.
1. IEBA	—																
2. rEBA	.37	—															
3. IFBA1	.30	.21	—														
4. rFBA1	.41 ^	.32	.30	—													
5. IFBA2	-.03	.36	-.02	.13	—												
6. rFBA2	.09	.24	.03	.55 **	.30	—											
7. IFFA1	.32	.23	.09	-.04	-.22	.01	—										
8. rFFA1	.32	.56 **	.17	.53 **	.31	.41 ^	.13	—									
9. IFFA2	-.18	.16	.19	.10	.45 ^	.08	.13	.39	—								
10. rFFA2	.03	.32	-.07	.37	.52 **	.58 ***	.05	.59 ***	.33	—							
11. lLOC	.31	.59 ***	.44 ^	.24	.28	-.02	.21	.42 ^	.43 ^	.12	—						
12. rLOC	.06	.46 *	.16	.01	.17	.00	.33	.26	.11	.35	.37	—					
13. IOFA	.13	.14	-.03	.08	.04	.27	.24	.31	.12	.24	.36	.39	—				
14. rOFA	.46 ^	.48 *	.50 *	.29	-.07	.04	.11	.32	.08	.07	.45 ^	.24	.12	—			
15. IMOA1	.01	.09	.22	.00	.29	.25	.28	.03	.35	.04	.27	.18	.28	-.11	—		
16. rMOA1	.04	.53 **	.05	.10	.44 ^	.32	.07	.63 ***	.36	.50 *	.51 *	.36	.57 ***	.23	.10	—	
17. IMOA2	.38	.52 **	.29	.14	.36	.21	.34	.44 ^	.17	.47 *	.40 ^	.21	.19	.34	.11	.50 *	—
18. rMOA2	.28	.54 **	.10	.17	.19	.23	.19	.31	.23	.10	.44 ^	.16	.20	.24	.30	.49 *	.45 ^

^BF₁₀ > 3, * BF₁₀ > 10, ** BF₁₀ > 30, *** BF₁₀ > 100

Table S3

The four patterns of change in correlation with ρ with increasing distance from ROI peak. Three sizes are shown per ROI: a sphere of 3mm radius (approximately 113mm³), a hollow ROI computed as a 9mm radius sphere minus a 6mm radius sphere (approximately 2148mm³), and a hollow ROI computed as a 15mm radius sphere minus a 12mm radius sphere (approximately 6896mm³). ROIs are grouped by pattern. Bold values have a $BF_{+0} > 3$.

Pattern Type	ROI	113 mm ³	2148 mm ³	6896 mm ³
A	rEBA	.49	.49	.48
A	lLOC	.49	.51	.47
A	rOFA	.47	.58	.59
A	rMOA1	.45	.48	.46
A	rLOC	.45	.48	.51
A	lEBA	.40	.51	.50
B	lOFA	.28	.48	.55
B	lMOA1	.06	.44	.46
B	lFFA1	.16	.36	.44
B	rFFA2	.24	.33	.36
B	lFBA1	.17	.34	.39
B	rFBA2	.17	.23	.36
C	rMOA2	.38	.37	.27
C	rFFA1	.38	.26	.25
D	lMOA2	.37	.14	.30
D	lFFA2	.10	.30	.23
D	lFBA2	.15	.20	.15
D	rFBA1	.07	.11	.18

Table S4

Anatomical ROIs, in descending order of the partial correlation of the residual neural scores correlation with α . The Pearson skipped correlation is also provided, as well as the mean signal in (and difference between) the two conditions, with standard errors.

ROI	Correlation with α		mean ALT	sd ALT	mean REP	sd REP	mean ALT-REP	SD ALT-REP
	Pearson	Pearson skipped						
Occipital_Mid_R	.53	.53	0.276	0.221	0.131	0.193	0.145	0.225
Occipital_Mid_L	.50	.50	0.200	0.238	0.081	0.187	0.119	0.209
Fusiform_R	.45	.45	0.198	0.187	0.061	0.119	0.137	0.209
Cuneus_R	.44	.49	-0.220	0.167	-0.089	0.220	-0.131	0.239
Occipital_Inf_L	.43	.36	0.481	0.311	0.219	0.220	0.261	0.267
Occipital_Sup_L	.42	.35	0.031	0.208	0.017	0.193	0.014	0.230
Occipital_Sup_R	.42	.39	0.088	0.212	0.040	0.185	0.048	0.222
Occipital_Inf_R	.41	.57	0.556	0.360	0.323	0.228	0.233	0.320
Fusiform_L	.39	.33	0.168	0.175	0.022	0.144	0.146	0.214
Parietal_Sup_R	.32	.28	0.042	0.200	0.034	0.165	0.008	0.279
Precentral_L	.32	.30	0.039	0.127	0.034	0.105	0.005	0.184
Lingual_R	.32	.28	-0.052	0.191	-0.008	0.179	-0.044	0.253
Parietal_Sup_L	.30	.30	0.071	0.176	-0.015	0.184	0.085	0.266
Parietal_Inf_R	.29	.31	0.092	0.171	0.044	0.130	0.048	0.237
Frontal_Inf_Oper_R	.28	.28	0.123	0.147	0.062	0.140	0.061	0.248
Cuneus_L	.25	.16	-0.342	0.166	-0.143	0.216	-0.199	0.263
Postcentral_L	.24	.26	-0.026	0.137	0.010	0.135	-0.036	0.212
Calcarine_R	.23	.26	-0.075	0.144	-0.022	0.170	-0.054	0.187
Frontal_Mid_R	.21	.24	-0.015	0.120	-0.002	0.134	-0.013	0.211
Precentral_R	.20	.22	-0.002	0.130	0.012	0.115	-0.014	0.178
Cingulum_Ant_R	.19	.23	-0.059	0.095	-0.021	0.083	-0.038	0.148
Temporal_Inf_L	.19	.06	0.041	0.161	-0.041	0.133	0.082	0.238
Lingual_L	.18	.35	-0.089	0.166	-0.019	0.157	-0.069	0.224
Insula_R	.18	.26	-0.021	0.098	0.012	0.086	-0.033	0.150
Supp_Motor_Area_R	.18	.36	-0.007	0.117	0.036	0.118	-0.044	0.189
Paracentral_Lobule_L	.16	.16	-0.051	0.144	-0.005	0.145	-0.046	0.202
Frontal_Inf_Tri_R	.16	.16	0.065	0.128	0.017	0.103	0.048	0.181
Temporal_Mid_R	.16	.23	-0.020	0.141	0.001	0.104	-0.021	0.167
Supp_Motor_Area_L	.14	.23	0.004	0.130	0.047	0.126	-0.043	0.208
Precuneus_R	.14	.18	-0.129	0.134	-0.074	0.186	-0.055	0.229
Frontal_Inf_Tri_L	.13	.14	0.009	0.097	-0.040	0.084	0.049	0.137
Precuneus_L	.10	.12	-0.150	0.142	-0.072	0.168	-0.078	0.222
Cingulum_Ant_L	.10	.10	-0.091	0.106	-0.024	0.099	-0.067	0.149
Temporal_Inf_R	.10	.06	0.173	0.187	0.084	0.134	0.089	0.250
Temporal_Sup_R	.10	.10	-0.080	0.103	0.001	0.103	-0.082	0.152
Frontal_Inf_Oper_L	.08	.06	0.055	0.127	-0.006	0.107	0.061	0.197
Temporal_Sup_L	.07	.07	-0.095	0.131	-0.008	0.108	-0.087	0.141
Calcarine_L	.07	.16	-0.100	0.187	-0.017	0.203	-0.083	0.274
Frontal_Mid_L	.05	.06	-0.046	0.081	-0.027	0.093	-0.019	0.151
Cingulum_Mid_R	.05	.27	-0.059	0.099	-0.005	0.107	-0.054	0.170
Frontal_Sup_Medial_L	.04	-.06	-0.107	0.135	-0.057	0.109	-0.051	0.177
Frontal_Sup_R	.03	-.01	-0.076	0.104	-0.019	0.106	-0.057	0.172
Temporal_Mid_L	.03	.02	-0.090	0.121	-0.026	0.115	-0.064	0.161
Parietal_Inf_L	.03	.04	0.050	0.133	-0.010	0.146	0.059	0.165
Postcentral_R	.03	.14	-0.013	0.134	0.017	0.115	-0.030	0.202
Cingulum_Mid_L	.02	.09	-0.090	0.101	-0.016	0.109	-0.074	0.162

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Table continued on next page.

Continuation of Table S4

SupraMarginal_R	.01	.01	0.004	0.128	0.046	0.104	-0.042	0.165
Paracentral_Lobule_R	.00	.04	-0.064	0.173	0.010	0.187	-0.073	0.306
Hippocampus_R	-.02	-.10	-0.047	0.102	-0.019	0.094	-0.028	0.142
Rolandic_Oper_R	-.03	-.02	-0.079	0.106	-0.004	0.108	-0.075	0.176
Hippocampus_L	-.04	-.03	-0.042	0.100	-0.008	0.082	-0.034	0.130
Heschl_L	-.04	.04	-0.073	0.115	-0.011	0.106	-0.063	0.149
Frontal_Sup_L	-.04	.19	-0.094	0.095	-0.041	0.083	-0.053	0.137
Angular_R	-.04	-.31	-0.050	0.171	-0.074	0.157	0.025	0.224
Frontal_Sup_Medial_R	-.04	-.15	-0.090	0.122	-0.050	0.114	-0.039	0.181
Insula_L	-.08	-.08	-0.023	0.096	0.010	0.077	-0.033	0.138
ParaHippocampal_R	-.08	.03	-0.071	0.137	-0.062	0.128	-0.009	0.214
Amygdala_L	-.10	-.08	-0.121	0.241	-0.014	0.185	-0.107	0.352
Rolandic_Oper_L	-.11	-.11	-0.059	0.121	0.014	0.118	-0.072	0.180
Heschl_R	-.12	-.12	-0.105	0.147	-0.013	0.109	-0.093	0.195
Amygdala_R	-.13	-.23	-0.096	0.221	-0.058	0.204	-0.038	0.316
ParaHippocampal_L	-.13	-.41	-0.088	0.142	-0.067	0.133	-0.021	0.216
SupraMarginal_L	-.14	-.16	-0.070	0.129	-0.015	0.132	-0.055	0.167
Cingulum_Post_R	-.17	-.15	-0.110	0.132	-0.027	0.125	-0.083	0.224
Cingulum_Post_L	-.19	-.13	-0.145	0.159	-0.040	0.128	-0.104	0.244
Angular_L	-.23	-.32	-0.185	0.185	-0.096	0.170	-0.089	0.228

Table S5

Whole brain clusters surviving an uncorrected threshold ($p < .01$) and cluster threshold ($k=3$ voxels).

Cluster	X	Y	Z
Left Postcentral	-36	-31	50
Left Middle Frontal	-24	41	29
Left Middle Cingulum	-15	-40	53
Right Precuneus	6	-64	-16
Right Frontal Inferior Oper	45	17	5

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